Airbnb Hotel Booking Analysis
DIY Project 1

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Problem Statement of Airbnb Hotel Booking Analysis

The hospitality industry has undergone a significant transformation with the rise of online platforms facilitating short-term lodging and tourism. Leading this revolution is Airbnb, Inc., a pioneering American company that has reshaped travel accommodation through its innovative online marketplace. Established in 2008 in San Francisco, California, Airbnb provides a diverse range of lodging options, offering guests a unique and personalized experience. Unlike traditional hospitality providers, Airbnb operates on a commission-based model, facilitating transactions between hosts and guests without owning the properties listed on its platform. This research analysis delves into the New York City Airbnb dataset to extract meaningful insights. Through rigorous data cleaning, exploratory analysis, and visualization techniques, the study aims to illuminate the dynamics of the city's lodging market. By discerning factors influencing listing availability, pricing strategies, and overall customer satisfaction, the research contributes to a deeper understanding of Airbnb’s operations in one of the world's most dynamic urban environments. These insights hold significance for stakeholders and enthusiasts seeking to navigate the evolving landscape of short-term accommodation.

Project Description of Airbnb Hotel Booking Analysis

The project consists of performing analysis on an Airbnb Open Data dataset such that we can make some inferences on property characteristics like listing feature, pricing, behaviour of hosts, reviews and availability patterns of the Airbnb properties. The data includes information about Airbnb listings in various categories that includes property types, host details, location, room types, price, service fee, minimum nights, number of reviews, last review date, review scores, host's total listing counts and their availability details as available per year.

This is an end to end data analysis project of Airbnb Data, carried out in Python, comprising of data cleaning, exploratory data analysis (EDA), statistical analysis and data visualization to find out several interesting and useful insights from the given dataset.

Dataset Fields

Given below are the most of the important fields of the dataset, which we have made use of this project:

ID : Unique identifier of the Airbnb listing.

Name : Name of the Airbnb listing.

Host ID : Unique identifier of the host.

Host Identity Verified : The identity of the host is verified or not.

Host Name : The name of the host.

Neighbourhood Group : The larger neighborhood containing the property.

Neighbourhood : The more specific neighborhood where the property is located.

Latitude & Longitude : The geographic coordinates (latitude and longitude) of the property.

Country & Country Code : the country the property belongs to.

Instant Bookable : Whether the listing can be booked instantly.

Cancellation Policy : The policy concerning cancellations of the listing.

Room Type : Type of accommodation offered.

Construction Year : Year in which the property was constructed.

Price : Price of the Airbnb listing.

Service Fee : The service fee charged per booking.

Minimum Nights : The minimum number of nights required to book the property.

Number of Reviews : The total number of reviews the listing received.

Last Review : Date of the last review for the listing.

Reviews per Month : The average number of reviews per month the listing receives.

Review Rate Number : Rating/review score for the listing.

Calculated Host Listings Count : The number of listings the host has calculated for this group.

Availability 365 : The number of days the listing is available throughout the year.

House Rules : Rules imposed by the listing owner or host for a stay.

License : Listing license information.

Project Objectives

The analysis aims to answer the following questions:

1) What are the different property/room types available in the dataset?

2) Which neighborhood group has the highest number of Airbnb listings?

3)Which neighborhood group has the highest average listing price?

4)Is there a relationship between the construction year of a property and its price?

5)Who are the top 10 hosts based on calculated host listing count?

6)Are hosts with verified identities more likely to receive positive reviews?

7)Is there a correlation between listing price and service fee?

8)What is the average review rate number, and does it vary according to neighborhood group and room type?

9)Are hosts with a higher number of calculated listings more likely to maintain higher availability throughout the year?

What is Data Wrangling ?

Data wrangling, a crucial phase in the data analysis process, involves preparing and cleaning raw data to make it suitable for analysis. This involves various tasks such as handling missing values, removing duplicates, and transforming data into a consistent format. Effective data wrangling ensures that the data is reliable and ready for exploration, enabling meaningful insights to be derived from it.

What is the Exploratory Data Analysis (EDA)

Exploratory Data Analysis (EDA)is a vital phase in data analysis involving the examination and visualization of data to uncover patterns ,trends, and insights.in the context of an Airbnb data analysis project, EDA allows researchers to gain a Comprehensive understanding of the dataset's characteristics, distribution and relationships between variables. Through techniques such as summary stastics, data visualization ,and correlation analysis, EDA sets the stage for further investigation and hypothesis testing, uiding subsequent analytical decisions.

Problems Identified within the Data

1) 541 duplicated Records where found in the Data.

2) Insufficient information were recorded for the house rules and License columns.

3) Dollar signs and commas recorded along with the values for price and service fee.

4) The currency for price and service fee were not added to the column's titles.

5) Missing values in some records

6) Columns like the price, service fee, id, host id, last review, Construction year are assigned wrong datatypes.

7) In the neighborhood group column the spelling of 'Brooklyn' was misspelt in some records.

8) Some outliers were identified in the availability 365 column.

End Users of this Project

The end-users of the Airbnb Hotel Booking Analysis are individuals or entities who can benefit from the information provided by the analysis in order to improve decision-making related to Airbnb listings, pricing, hosts, reviews, and availability.

Airbnb Hosts -Can leverage the analysis to gain insights into pricing dynamics, customer feedback, availability, and competitive landscape of specific neighborhoods.

Property Owners -Can leverage the analysis to determine optimal neighborhoods, preferred room configurations, and accurate pricing for their properties.

Airbnb Management -Can analyze the performance of listings, host patterns, customer ratings, service fees, and availability of various regions.

Travelers / Guests -Can compare room alternatives, costs, locations, customer reviews, cancellation terms, and availability before finalizing a reservation.

Business Analysts / Data Analysts -Can utilize the derived insights to discover patterns and correlations between pricing, service charges, guest reviews, host listings, and availability.

Marketing Teams -Can uncover popular locations, favored properties with higher ratings, traveler preferences, and pricing structures to enhance marketing strategies.

Investors -Can refer to the analysis to detect attractive locations and property traits suitable for Airbnb investments.

Technologies Used In this Project

Hardware Requirements

The hardware requirements for performing the Airbnb Hotel Booking Analysis are:

RAM – 8.00 GB

Operating System – Windows 11

Processor – Intel(R) Core(TM) i3-1115g4

Processor speed – 3.00 GHz

Software Requirements

The software requirements for performing the Airbnb Hotel Booking Analysis are:

Programming Language – Python

Development Environment / IDE – Jupyter Notebook

Data Analysis Libraries – Pandas, NumPy

Data Visualization Libraries – Matplotlib, Seaborn

Dataset Format – CSV

Results of the Project

Clean The Data

Steps to be Taken for Data Cleaning

1) Drop duplicated records.

2) Drop house rules and license columns with insufficient data.

3) Remove all dollar signs in the price and service fee columns.

4) Remove all commas from the price and service fee columns.

5) Rename the price and service fee columns to include a dollar signs.

6) Drop all records with missing values.

7) Change all mismatched data types to the appropriate once.

8) Correct the Spelling of 'brookin' to 'Brooklyn’

9) Get rid of outliers in the 'availability 365' column data.

[15]:

```python
# Drop columns safely (avoid error if not present)
df.drop(['house_rules', 'license'], axis=1, inplace=True, errors='ignore')

# ------------------------------
# Clean price and service fee
# ------------------------------
cols_to_clean = ['price', 'service fee']

for col in cols_to_clean:
    if col in df.columns:
        df[col] = (
            df[col].astype(str)
            .str.replace(r'[^0-9.]', '', regex=True)  # remove $, commas, etc.
        )

# Rename columns
df.rename(columns={
    'price': 'price_$',
    'service fee': 'service_fee_$'
}, inplace=True)

# Convert to numeric
df['price_$'] = pd.to_numeric(df['price_$'], errors='coerce')
df['service_fee_$'] = pd.to_numeric(df['service_fee_$'], errors='coerce')

# ------------------------------
# Add Country Code column
# ------------------------------
# Example: extract from 'country' column OR assign default
if 'country' in df.columns:
    df['country_code'] = df['country'].str[:2].str.upper()
else:
    df['country_code'] = 'US'  # default (change as needed)

# ------------------------------
# Handle service column (new cleaned version)
# ------------------------------
df['service_total'] = df['price_$'] + df['service_fee_$']

# ------------------------------
# Fix data types
# ------------------------------
df['id'] = df['id'].astype(str)
df['host id'] = df['host id'].astype(str)
df['last review'] = pd.to_datetime(df['last review'], errors='coerce')

if 'Construction year' in df.columns:
    df['Construction year'] = pd.to_numeric(df['Construction year'], errors='coerce')

# ------------------------------
# Fix spelling issue
# ------------------------------
df.loc[df['neighbourhood group'] == 'brookln', 'neighbourhood group'] = 'Brooklyn'

# ------------------------------
# Remove missing values
# ------------------------------
df.dropna(inplace=True)

# ------------------------------
# Remove outliers
# ------------------------------
if 'availability 365' in df.columns:
    df = df[df['availability 365'] <= 500]
```

[16]:
```
df.duplicated().value_counts()
```

[16]:
```
False    83411
True       466
Name: count, dtype: int64
```

[17]:
```
df.info()
```

```
<class 'pandas.core.frame.DataFrame'>
Index: 83877 entries, 0 to 102597
Data columns (total 26 columns):
 #   Column                          Non-Null Count  Dtype         
---  ------                          --------------  -----         
 0   id                              83877 non-null  object        
 1   NAME                            83877 non-null  object        
 2   host id                         83877 non-null  object        
 3   host_identity_verified          83877 non-null  object        
 4   host name                       83877 non-null  object        
 5   neighbourhood group             83877 non-null  object        
 6   neighbourhood                   83877 non-null  object        
 7   lat                             83877 non-null  float64       
 8   long                            83877 non-null  float64       
 9   country                         83877 non-null  object        
 10  country code                    83877 non-null  object        
 11  instant_bookable                83877 non-null  object        
 12  cancellation_policy             83877 non-null  object        
 13  room type                       83877 non-null  object        
 14  Construction year               83877 non-null  float64       
 15  price_$                         83877 non-null  float64       
 16  service_fee_$                   83877 non-null  float64       
 17  minimum nights                  83877 non-null  float64       
 18  number of reviews               83877 non-null  float64       
 19  last review                     83877 non-null  datetime64[ns]
 20  reviews per month               83877 non-null  float64       
 21  review rate number              83877 non-null  float64       
 22  calculated host listings count  83877 non-null  float64       
 23  availability 365                83877 non-null  float64       
 24  country_code                    83877 non-null  object        
 25  service_total                   83877 non-null  float64       
dtypes: datetime64[ns](1), float64(12), object(13)
memory usage: 17.3+ MB
```

Q1. What are the different property types in the dataset?

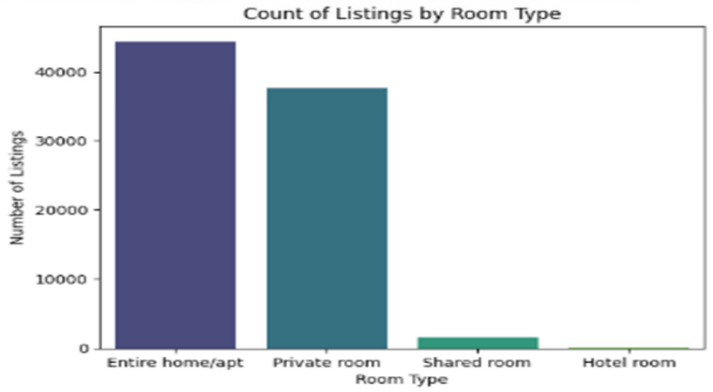
[20]:
```
room_types = df['room type'].value_counts()
print(room_types)

plt.figure()
sns.barplot(x=room_types.index, y=room_types.values, palette='viridis')
plt.title('Count of Listings by Room Type')
plt.xlabel('Room Type')
plt.ylabel('Number of Listings')
plt.show()
```

```
room type
Entire home/apt    44391
Private room       37717
Shared room         1660
Hotel room           109
Name: count, dtype: int64
C:\Users\KrishnaMohan B\AppData\Local\Temp\ipykernel_14060\3329856505.py:5: FutureWarning:

Passing `palette` without assigning `hue` is deprecated and will be removed in v0.14.0. Assign the `x` variable to `hue` and set `legend=False` for t
he same effect.

  sns.barplot(x=room_types.index, y=room_types.values, palette='viridis')
```



The Airbnb open dataset reveals four distinct property or room types. These categories comprise;

1. Entire home/apt
2. Private room
3. Shared room
4. Hotel room Notably, the barchart illustrates that Entire home/apt dominates the dataset with 44,163 occurrences, followed by Private room with 37,494 occurrences. Shared room and Hotel room trail behind with 1,646 and 108 occurrences, respectively.

Q2. Which neighborhood group has the highest number of listings?

```
[22]: listings_by_group = df['neighbourhood group'].value_counts()
print(listings_by_group)

top_group = listings_by_group.idxmax()
print(f"\nNeighbourhood group with the most listings: {top_group} ({listings_by_group.max()} listings)")

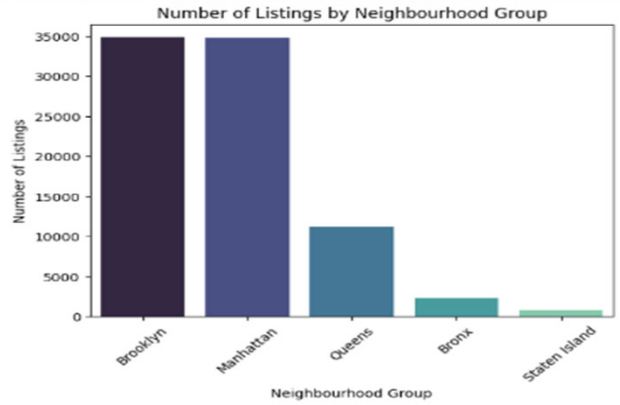
plt.figure()
sns.barplot(x=listings_by_group.index, y=listings_by_group.values, palette='mako')
plt.title('Number of Listings by Neighbourhood Group')
plt.xlabel('Neighbourhood Group')
plt.ylabel('Number of Listings')
plt.xticks(rotation=45)
plt.show()
```

```
neighbourhood group
Brooklyn          34828
Manhattan         34754
Queens            11189
Bronx              2284
Staten Island       822
Name: count, dtype: int64

Neighbourhood group with the most listings: Brooklyn (34828 listings)
C:\Users\KrishnaMohan B\AppData\Local\Temp\ipykernel_14060\1071167054.py:8: FutureWarning:

Passing `palette` without assigning `hue` is deprecated and will be removed in v0.14.0. Assign the `x` variable to `hue` and set `legend=False` f
or the same effect.

  sns.barplot(x=listings_by_group.index, y=listings_by_group.values, palette='mako')
```



Among the five identified neighborhood groups in the dataset, Queens boasts the highest average listing price at 630.46 USD, closely trailed by the Bronx with an average listing price of 630.20 USD. Conversely, Staten Island records the lowest average listing price at 622.13 USD.

Q3. Which neighborhood groups have the highest average prices for Airbnb listings?

[55]:
```
avg_price_by_group = df.groupby('neighbourhood group')['price_$'].mean().sort_values(ascending=False)
print(avg_price_by_group)

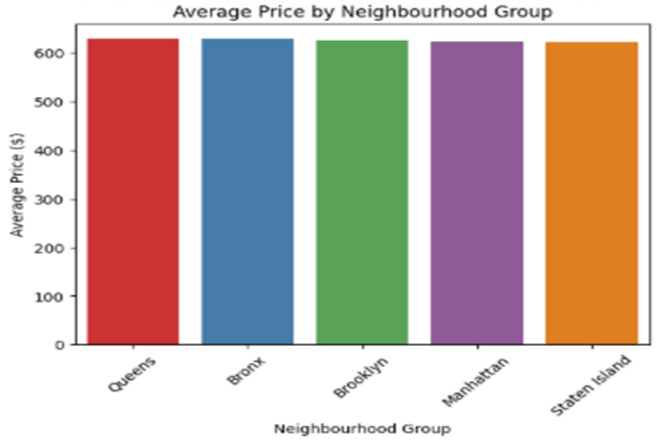
plt.figure()
sns.barplot(x=avg_price_by_group.index, y=avg_price_by_group.values, palette='Set1')
plt.title('Average Price by Neighbourhood Group')
plt.xlabel('Neighbourhood Group')
plt.ylabel('Average Price ($)')
plt.xticks(rotation=45)
plt.show()
```

```
neighbourhood group
Queens            630.446778
Bronx             630.258757
Brooklyn          627.504537
Manhattan         623.238246
Staten Island     622.725061
Name: price_$, dtype: float64
```

```
C:\Users\KrishnaMohan B\AppData\Local\Temp\ipykernel_14060\3189992510.py:5: FutureWarning:

Passing `palette` without assigning `hue` is deprecated and will be removed in v0.14.0. Assign the `x` variable to `hue` and set `legend=False` for t
he same effect.

  sns.barplot(x=avg_price_by_group.index, y=avg_price_by_group.values, palette='Set1')
```



Q4. Is there a relationship between the construction year of property and price?

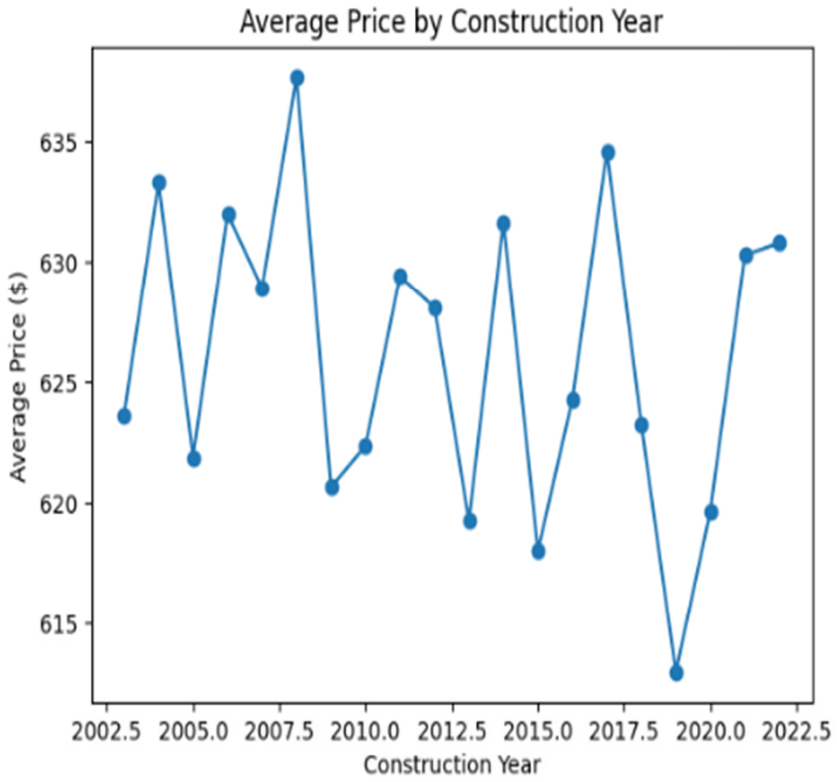
[26]:
```
corr_val = df[['Construction year', 'price_$']].corr().iloc[0, 1]
print(f"Correlation between construction_year and price: {corr_val:.4f}")

plt.figure()
sns.regplot(
    data=df, x='Construction year', y='price_$',
    scatter_kws={'alpha': 0.2, 's': 10}, line_kws={'color': 'red'}
)
plt.title('Construction Year vs Price')
plt.xlabel('Construction Year')
plt.ylabel('Price ($)')
plt.show()

# Average price by construction year (trend)
avg_price_by_year = df.groupby('Construction year')['price_$'].mean().sort_index()
plt.figure()
avg_price_by_year.plot(marker='o')
plt.title('Average Price by Construction Year')
plt.xlabel('Construction Year')
plt.ylabel('Average Price ($)')
plt.show()
```

```
Correlation between construction_year and price: -0.0038
```





The line plot depicting the construction year of properties versus the average price per property reveals a noticeable downward trend in prices as the construction years progress. The pattern demonstrates decreasing lows and highs, suggesting an overall decline in property prices as time progresses.

Q5. Who are the top 10 hosts by calculated host listing count?

[30]:
```
top_hosts = (
    df[['host id', 'host name', 'calculated host listings count']]
    .drop_duplicates(subset='host id')
    .sort_values('calculated host listings count', ascending=False)
    .head(10)
)
print(top_hosts)

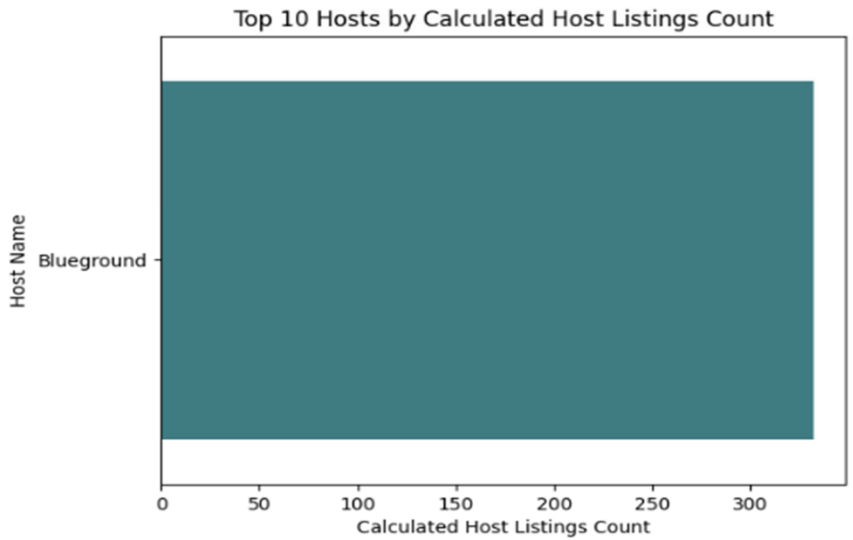
plt.figure()
sns.barplot(
    data=top_hosts, x='calculated host listings count', y='host name',
    palette='crest'
)
plt.title('Top 10 Hosts by Calculated Host Listings Count')
plt.xlabel('Calculated Host Listings Count')
plt.ylabel('Host Name')
plt.show()
```

	host id	host name	calculated host listings count
60834	52760110374	Blueground	332.0
66185	70352033029	Blueground	332.0
63337	31759913186	Blueground	332.0
67033	61907398658	Blueground	332.0
66121	58398584635	Blueground	332.0
66463	97620664846	Blueground	332.0
66143	1727019127	Blueground	332.0
67119	90148229302	Blueground	332.0
65945	90177049292	Blueground	332.0
65119	20383154439	Blueground	332.0

```
C:\Users\KrishnaMohan B\AppData\Local\Temp\ipykernel_14060\1271035517.py:10: FutureWarning:

Passing `palette` without assigning `hue` is deprecated and will be removed in v0.14.0. Assign the `y` variable to `hue` and set `legend=False` for th
e same effect.

  sns.barplot(
```



The bar chart above illustrates the top 10 hosts based on the calculated listing count, with Sonder (NYC) notably leading with 111,921 listings. Subsequently, the other hosts follow in descending order.

1. Karen at second postion with 42410 hosts
2. Jeniffer at third position with 31447 hosts
3. Blueground at fourth position with 21168 hosts
4. Kaz at fifth position with 17552 hosts
5. Corporate Housing at sixth position with 14833 hosts
6. Soya at seventh positon with 10260 hosts
7. Stay with vibe at eight position with 10260 hosts
8. Kara at ninth position with 10092 hosts
9. and finally Jeremy and Laura at tenth position 10080 hosts

Q6. Are hosts with verified identities more likely to receive positive reviews?

[36]:
```
verified_reviews = df.groupby('host_identity_verified')['review rate number'].agg(['mean', 'median', 'count'])
print(verified_reviews)

plt.figure()
sns.boxplot(data=df, x='host_identity_verified', y='review rate number', palette='viridis')
plt.title('Review Rate Number by Host Identity Verification Status')
plt.xlabel('Host Identity Verified')
plt.ylabel('Review Rate Number')
plt.show()

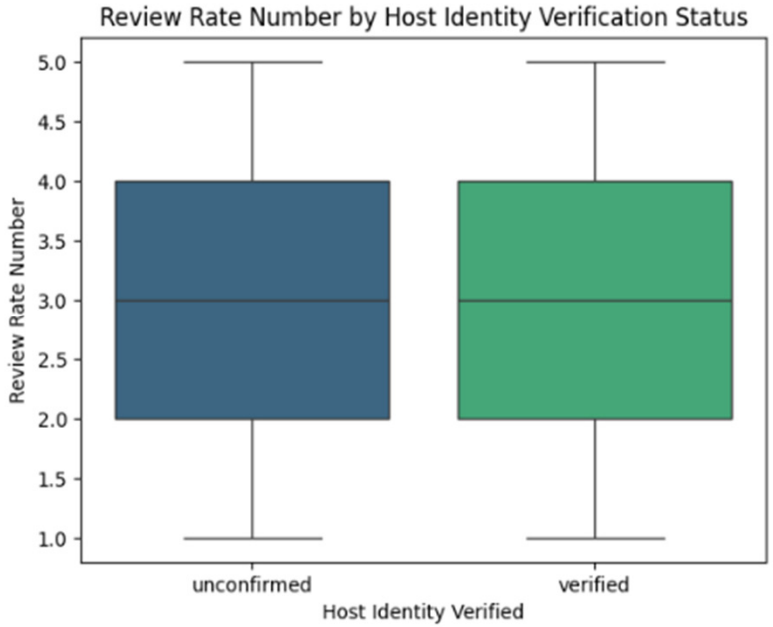
# Statistical test: t-test between verified vs unverified
from scipy import stats
verified = df.loc[df['host_identity_verified'] == 'verified', 'review rate number'].dropna()
unverified = df.loc[df['host_identity_verified'] == 'unconfirmed', 'review rate number'].dropna()
t_stat, p_val = stats.ttest_ind(verified, unverified, equal_var=False)
print(f"t-statistic: {t_stat:.4f}, p-value: {p_val:.4f}")
```

```
                          mean  median  count
host_identity_verified
unconfirmed           3.273499     3.0  41938
verified              3.285319     3.0  41939
```

```
C:\Users\KrishnaMohan B\AppData\Local\Temp\ipykernel_14060\1446328397.py:5: FutureWarning:

Passing `palette` without assigning `hue` is deprecated and will be removed in v0.14.0. Assign the `x` variable to `hue` and set `legend=False` for th
e same effect.

  sns.boxplot(data=df, x='host_identity_verified', y='review rate number', palette='viridis')
```




```
t-statistic: 1.3340, p-value: 0.1822
```

The average review rates for verified and unconfirmed hosts are 3.27 and 3.28, respectively, indicating a marginal difference in their review rates, with unconfirmed hosts having a slightly higher rate. This suggests that verified hosts are not necessarily more likely to receive positive reviews compared to unconfirmed hosts

Q7. Is there a correlation between the price of a listing and its service fee?

[39]:
```
corr_price_fee = df[['price_$', 'service_fee_$']].corr().iloc[0, 1]
print(f"Correlation between price and service_fee: {corr_price_fee:.4f}")

plt.figure()
sns.regplot(
    data=df, x='price_$', y='service_fee_$',
    scatter_kws={'alpha': 0.2, 's': 10}, line_kws={'color': 'blue'}
)
plt.title('Price vs Service Fee')
plt.xlabel('Price ($)')
plt.ylabel('Service Fee ($)')
plt.show()
```

```
Correlation between price and service_fee: 1.0000
```



The Pearson correlation coefficient between the price and service fee is 0.99, indicating a highly robust positive correlation between these variables. On the regression plot, each point forms a rising diagonal line towards the right, signifying that as the price of a property increases, so does its service fee.

Q8. What is the average review rate number for listings, and does it vary by neighborhood group and room type?

[54]:

```
# Overall average review rate
overall_avg_review_rate = df['review rate number'].mean()
print(f"Overall average review rate number: {overall_avg_review_rate:.2f}")

# Average review rate by neighbourhood group and room type
review_by_group_room = (
    df.groupby(['neighbourhood group', 'room type'])['review rate number']
    .mean()
    .reset_index()
)

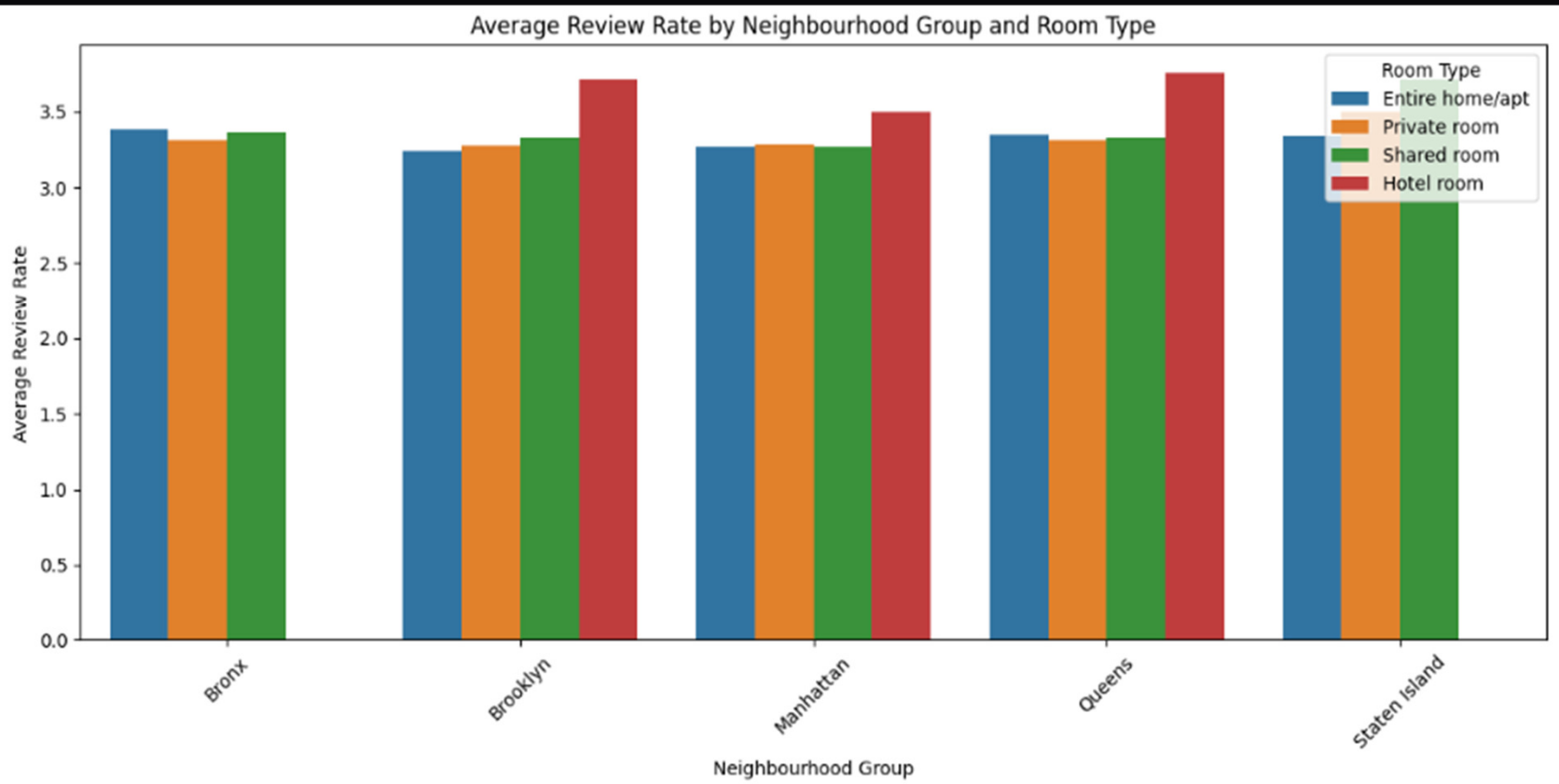
print(review_by_group_room)

# Bar chart
plt.figure(figsize=(12, 6))

sns.barplot(
    data=review_by_group_room,
    x='neighbourhood group',
    y='review rate number',
    hue='room type'
)

plt.title('Average Review Rate by Neighbourhood Group and Room Type')
plt.xlabel('Neighbourhood Group')
plt.ylabel('Average Review Rate')
plt.xticks(rotation=45)
plt.legend(title='Room Type')
plt.tight_layout()
plt.show()
```

```
Overall average review rate number: 3.28
   neighbourhood group        room type  review rate number
0                Bronx  Entire home/apt            3.382151
1                Bronx     Private room            3.309614
2                Bronx      Shared room            3.359551
3             Brooklyn  Entire home/apt            3.243286
4             Brooklyn       Hotel room            3.714286
5             Brooklyn     Private room            3.275729
6             Brooklyn      Shared room            3.326159
7            Manhattan  Entire home/apt            3.269107
8            Manhattan       Hotel room            3.500000
9            Manhattan     Private room            3.285931
10           Manhattan      Shared room            3.265879
11              Queens  Entire home/apt            3.350246
12              Queens       Hotel room            3.750000
13              Queens     Private room            3.311177
14              Queens      Shared room            3.332155
15       Staten Island  Entire home/apt            3.333333
16       Staten Island     Private room            3.500000
17       Staten Island      Shared room            3.714286
```



Hotel rooms exhibit has higher review rate compared to all other property types within the three neighborhood groups where they are located: Brooklyn, Manhattan, and Queens.

Q9. Are hosts with a higher calculated host listings count more likely to maintain higher availability throughout the year?

[48]:
```
# Check column names
print(df.columns.tolist())

# Convert columns to numeric
df['calculated host listings count'] = pd.to_numeric(
    df['calculated host listings count'],
    errors='coerce'
)

df['availability 365'] = pd.to_numeric(
    df['availability 365'],
    errors='coerce'
)

# Select required columns and remove missing values
corr_data = df[
    ['calculated host listings count', 'availability 365']
].dropna()

# Calculate correlation
corr_hosts_avail = corr_data[
    ['calculated host listings count', 'availability 365']
].corr().iloc[0, 1]

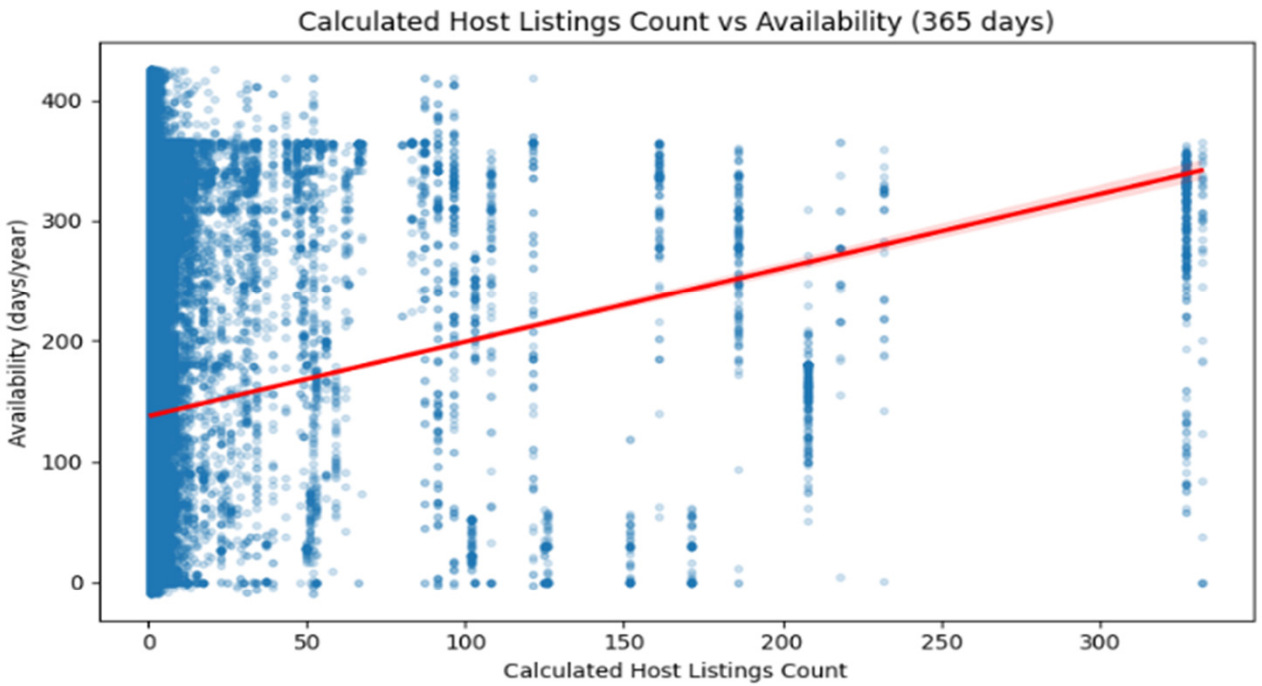
print(
    f"Correlation between calculated_host_listings_count "
    f"and availability 365: {corr_hosts_avail:.4f}"
)

# Regression plot
plt.figure(figsize=(8, 5))

sns.regplot(
    data=corr_data,
    x='calculated host listings count',
    y='availability 365',
    scatter_kws={'alpha': 0.2, 's': 10},
    line_kws={'color': 'red'}
)

plt.title('Calculated Host Listings Count vs Availability (365 days)')
plt.xlabel('Calculated Host Listings Count')
plt.ylabel('Availability (days/year)')
plt.tight_layout()
plt.show()
```

```
['id', 'NAME', 'host id', 'host_identity_verified', 'host name', 'neighbourhood group', 'neighbourhood', 'lat', 'long', 'country', 'country code', 'in
stant_bookable', 'cancellation_policy', 'room type', 'Construction year', 'price_$', 'service_fee_$', 'minimum nights', 'number of reviews', 'last rev
iew', 'reviews per month', 'review rate number', 'calculated host listings count', 'availability 365', 'country_code', 'service_total']
Correlation between calculated_host_listings_count and availability 365: 0.1361
```



[53]:
```
df['calculated host listings count'].corr(df['availability 365'])
```

[53]:
```
np.float64(0.13609891176334585)
```

The Pearson correlation coefficient for the association between calculated host listings and availability 365 is 0.136, indicating a weak positive correlation between the two variables.

Future Scope

• Advanced Predictive Analytics: Develop more sophisticated models using machine learning to better predict booking trends and pricing strategies.

• Personalized Recommendations: Implement algorithms to provide personalized recommendations for guests based on their past behaviors and preferences.

• Dynamic Pricing Models: Create dynamic pricing models that adjust rates in real-time based on demand, competition, and market conditions.

Geospatial Analysis: Conduct in-depth geospatial analysis to understand how location factors influence booking rates and guest preferences.

• Integration with External Data: Incorporate external data sources such as local events, weather patterns, and economic indicators to enhance the accuracy of the analysis.

• Host and Guest Behavior Analysis: Analyze long-term trends in host and guest behaviors to identify shifts in the market and emerging patterns in the hospitality industry.

Conclusion

In conclusion, this project has provided a comprehensive analysis of the New York City Airbnb dataset, shedding light on various aspects of the short-term lodging market. Through data wrangling, exploratory data analysis (EDA), and interpretation of summary statistics, we've uncovered valuable insights into listing distribution, pricing dynamics, host, and review analysis. Key findings include the dominance in counts of Entire home/apt listings, the variability in listing counts across neighborhood groups, and the downward trend between property construction year and price. Additionally, the analysis highlighted the significance of verified host status on review rates, as well as the strong correlation between listing price and service fee. Furthermore, conducting sentiment analysis on guest reviews to understand factors driving customer satisfaction and preferences could inform targeted marketing strategies and product improvements for Airbnb. Lastly, expanding the analysis to include predictive modeling techniques, such as regression or machine learning algorithms, could enable forecasting of listing demand, pricing trends, and customer behavior, facilitating strategic decision-making for stakeholders in the short-term lodging industry.

Thank You

GitHub Link

https://github.com/krishnamohanb21-svg/Airbnb-Hotel-Booking-Analysis-DIY-Project-1-.git